SIGNBRIDGE
REAL-TIME INDIAN SIGN LANGUAGE
RECOGNITION IN THE BROWSER

PROJECT REPORT

Human-Computer Interaction (HCI) Continuous Evaluation Assignment

On-device recognition of static ISL fingerspelling (A–Z, 1–9) with MediaPipe hand landmarks and a neural-network classifier

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Abbreviations

Abbreviation	Meaning
HCI	Human–Computer Interaction
ISL	Indian Sign Language
ML	Machine learning
MLP	Multilayer perceptron (feed-forward neural network)
RBF / SVM	Radial basis function / Support vector machine
kNN	k-nearest neighbours
WASM	WebAssembly
UI	User interface

1. Abstract

SignBridge is a web application that recognises static Indian Sign Language (ISL) signs from a webcam and shows the meaning as large text and speech. It covers the 35 static fingerspelling signs A–Z and 1–9. Each video frame goes through MediaPipe Hand Landmarker, which finds up to two hands and 21 key points per hand. The points are normalised into an 85-value feature vector and classified by a small multilayer perceptron (85-128-64-35). The model was trained with scikit-learn on a public dataset of 42,745 ISL images. The 358 duplicated files were removed, and landmarks were found in 98.79% of the remaining images. On a held-out test set of 8,377 images the MLP reached **97.86% accuracy** (macro-F1 97.42%). The split was designed so that near-duplicate video frames cannot appear in both training and test data. Inference runs fully in the browser, so camera frames never leave the user's device.

The interface was designed through a user-centred process: identification of user groups, a ten-question survey of 10 respondents, personas, a literature review and research-gap analysis, then low-fidelity wireframes and a working prototype. All eight of Shneiderman's Golden Rules are applied and documented. Examples are keyboard shortcuts, live feedback with a hand skeleton and confidence bar, a confidence threshold that shows an "uncertain" state instead of a wrong answer, and Undo for every saving or deleting action.

Keywords: Indian Sign Language, sign language recognition, MediaPipe, hand landmarks, multilayer perceptron, on-device inference, accessibility, human-computer interaction, Shneiderman's golden rules, wireframes.

2. Introduction

2.1 Purpose

The purpose of the system is to help a person who signs ISL and a person who does not know ISL communicate short, exact information such as names, spellings, numbers and codes. The camera recognises fingerspelled letters and numbers and shows them as text and speech. The same tool helps learners practise ISL fingerspelling against reference photos.

2.2 Scope

The system recognises the 35 static ISL signs A–Z and 1–9 from a live webcam, one sign at a time. It includes the dataset preparation and model-training pipeline, the in-browser recognition pipeline, result confirmation and correction, history, a reference vocabulary and accessibility settings. It does not translate sentences or recognise word signs that need movement or facial expression. It is not a certified interpreter.

2.3 Objectives

- Train a lightweight classifier for static ISL signs on hand-landmark features and measure its accuracy honestly on held-out data.
- Run detection and classification on the user's device in real time, without uploading video.
- Present results so that users can see how sure the system is, and correct, undo or retry easily.
- Design the interface around the needs of identified user groups, using a user survey, personas, wireframes and Shneiderman's Eight Golden Rules.

2.4 Problem statement

Deaf and hard-of-hearing people who use ISL often meet hearing people who do not know any ISL, such as shopkeepers, office staff and classmates. Without an interpreter, exact information like a name, a phone number or a room number is slow to pass on and easy to get wrong. Existing research systems report recognition accuracy but rarely ask how the result should be shown to users, or what should happen when the system is wrong.

2.5 Research question

How can a camera-based ISL recogniser present its output so that both signers and non-signers can trust it, notice when it is unsure, and correct it quickly?

3. Identification of User Groups

3.1 Problem relevance and context

ISL is the main language of many Deaf people in India, while very few hearing people know it. Fingerspelling is used for names, places, numbers and words that have no sign. These are exactly the items that are hardest to guess from gestures or lip reading. A phone or laptop camera is now available almost everywhere, so a tool that turns fingerspelling into text could be used at a shop counter, in a classroom or at home.

3.2 Target user groups

User group	Who they are	Characteristics	Needs	Context of use
ISL users (primary)	Deaf or hard-of-hearing people who sign ISL	Fluent in ISL; varying reading skill in English; experienced with phones; used to being misunderstood	Be understood quickly when spelling names or numbers; see that the system read the sign correctly; not depend on another person	Shops, offices, hospitals, college; mostly phone or laptop; often in a hurry
Hearing communication partners (primary)	Family members, classmates, teachers, shop and office staff	Know no ISL or only a few signs; no technical background	A large, clear text (and spoken) answer; know when the answer may be wrong; very few steps	Counter or desk, laptop or shared screen; short interactions
ISL learners (secondary)	Students and trainees learning fingerspelling	Motivated, comfortable with apps; unsure about hand shapes	Practise one sign at a time against a reference; immediate feedback on whether it matched	At home or in class on a laptop
Technical evaluators (secondary)	CS students, developers, instructors	Comfortable with ML concepts	See confidence, alternatives and how the model works; reproduce the results	Laptop at a desk

The system is not intended for legal, medical or emergency communication, where a qualified interpreter is required.

4. User Survey and Questionnaire

4.1 Method

A ten-question questionnaire was prepared as a Google Form ("User Survey on SignBridge: Indian Sign Language Detection") and completed in person, on a phone, by 10 respondents on 25 October 2026 (Section 4.5). Every respondent answered the same questions (Section 4.2), so answers can be counted and compared. Each response is kept with its ID (R01–R10) in `docs/interviews.md` and the raw export in `docs/survey_responses.csv`. The answers were grouped into eight findings (Section 4.4), and each finding was linked to a design decision in the prototype.

4.2 Questionnaire

No.	Question	Response options	What it tells us
Q1	Which category best describes you?	Deaf or hard of hearing; ISL user or learner; Hearing person who does not know ISL	User group
Q2	How familiar are you with Indian Sign Language (ISL)?	Very familiar; Somewhat familiar; Know a few signs; Have heard of it but know no signs; Not familiar at all	Expertise
Q3	How often do you interact with people who use sign language?	Daily; Monthly; Rarely; …	Relevance, context
Q4	How difficult do you think communication can be between an ISL user and someone who does not know ISL?	Not difficult … Extremely difficult	Problem relevance
Q5	Have you ever used a sign language recognition or translation application?	Yes, regularly; I have tried a similar tool; No, but I know about them; No, never	Current practice
Q6	How useful would a system that recognises ISL signs through a camera and displays the corresponding words be?	1 (not useful) – 5 (very useful)	Perceived value
Q7	Which output would be most useful to you?	Text on screen; Text-to-speech audio; Both text and speech; Visual sign demonstration	Output and feedback needs
Q8	Which feature would be most important in such an application?	High recognition accuracy; Real-time results; Simple and accessible interface; Ability to correct wrong predictions; Privacy and camera control	Priorities
Q9	What should the application do when it is unsure about a sign?	Show a message that the sign is uncertain; Ask the user to repeat the sign; Offer possible alternative words; Allow the user to type the intended message	Error handling, trust
Q10	How likely would you be to use or recommend such a system if it worked reliably?	Very likely; Likely; Not sure; Unlikely; …	Adoption

4.3 Respondents

Group (Q1)	Count	IDs

Hearing person who does not know ISL	5	R01, R03, R05, R07, R10
ISL user or learner	3	R02, R04, R06
Deaf or hard of hearing	2	R08, R09

Familiarity with ISL (Q2): very familiar 1, somewhat familiar 1, know a few signs 2, heard of it but know no signs 2, not familiar at all 4. Contact with signers (Q3): daily 1, monthly 5, rarely 4. Some self-descriptions do not match the familiarity answer (for example, two "ISL user or learner" respondents and one "Deaf or hard of hearing" respondent report knowing no signs), so the group labels are treated as indicative only (Section 16).

4.4 Key findings

No.	Finding	What respondents said (count of 10)	How the prototype responds
F1	The problem is real	Q4: 8 rated communication very (4) or extremely (4) difficult; 1 slightly, 1 not difficult. Q3: 6 meet signers monthly or daily.	Core task T1: recognise a sign and show it to the hearing partner in one step.
F2	Text and speech together	Q7: both text and speech 5, speech only 2, text only 2, visual sign demonstration 1. So 7 want speech and 7 want text.	Large text result plus a Speak button (S); reference photo of the sign for the one who wanted a visual.
F3	Correcting mistakes matters most	Q8: ability to correct wrong predictions was the most chosen feature (3).	Incorrect (X) with the intended sign, Try again (R), Undo (U) on every save or delete.
F4	Simple, fast and private	Q8: simple and accessible interface 2, real-time results 2, privacy and camera control 2, high accuracy 1.	One main action per screen, live feedback about 12 times a second, camera only on request, all processing on the device.
F5	Never guess silently when unsure	Q9: ask to repeat 4, let me type the message 3, show an "uncertain" message 2, offer alternatives 1. Nobody chose a silent best guess.	Below 60% confidence: uncertain state with Try again, closest guesses and a box to type the message and speak it.
F6	Most have never used such a tool	Q5: 6 have never used one (2 never heard of them, 4 know of them); 3 tried a similar tool; 1 uses one regularly.	Guided first screen, hints and key labels on buttons, How It Works page, Demo mode without a camera.
F7	Value must be earned by reliability	Q6: mean usefulness 3.0 / 5 (only 3 gave 4 or 5). Q10: 5 likely or very likely to use it if reliable, 4 not sure, 1 unlikely.	Honest scope banner, tested accuracy shown in the app, confidence on every result, uncertain state instead of wrong answers.
F8	The experienced signer cares about accuracy	R09 (Deaf, very familiar with ISL, daily contact, uses apps regularly) rated usefulness 1, chose high accuracy and an "uncertain" message, yet would be very likely to use a reliable system.	Confidence threshold, 9-of-12 frame agreement, honest leak-free accuracy reporting (Section 10).

4.5 Evidence

All responses are kept unedited in `docs/survey_responses.csv` (exported from Google Forms with timestamps) and tabulated in `docs/interviews.md`. Photos were taken with the respondents' permission while they filled in the form.



Figures 4.1–4.6: in-person survey sessions (photos).

5. User Persona Development

Three personas were built from the survey responses, one for each user group. Each is a composite: the name and age are illustrative, but every goal, need, pain point and behaviour comes from the responses listed under "Based on".

Persona 1: Meera, 24, Deaf ISL signer	
User group	Deaf or hard of hearing; fluent ISL user
Goal	Get names, numbers and short words across to hearing people without depending on an interpreter.
Needs	High recognition accuracy; clear text on screen; an honest "uncertain" message instead of a wrong word.
Pain point	Finds communication extremely difficult; existing apps are not reliable enough, so she rates their usefulness low.
Behaviour	Talks with hearing people daily; already uses sign language apps regularly; would very likely use a tool that works reliably.
Usage context	Phone or laptop at a counter, office or college.
Based on	R09, R08; findings F1, F5, F7, F8

Persona 2: Rohit, 21, hearing classmate who does not know ISL	
User group	Hearing person who does not know ISL
Goal	Understand a signing classmate or customer quickly during short conversations.
Needs	Both text and speech; real-time results; a way to correct wrong predictions; when unsure, ask to repeat the sign.
Pain point	Knows no or very few signs and thinks communication is very or extremely difficult; has never used a recognition app.

Behaviour	Meets signers monthly or rarely; likely to use the app if it is reliable.
Usage context	Laptop or phone in class or at work; quick, one-off interactions.
Based on	R05, R07, R10, R01; findings F1, F2, F3, F4, F5, F6

Persona 3: Sana, 20, ISL learner	
User group	ISL user or learner, still at an early stage
Goal	Practise signs and be understood while she is still learning.
Needs	Speech output; a simple, accessible interface; privacy and control over the camera; a way to type the message when the sign is not recognised.
Pain point	Unsure about hand shapes; has tried a similar tool before but is not sure she would keep using one.
Behaviour	Meets signers monthly; rates the idea 3–4 out of 5.
Usage context	At home or in class on a laptop.
Based on	R02, R04, R06; findings F2, F4, F5, F6, F7

6. Literature Review

Six sources were reviewed. Two cover sign language recognition as a field, two are specific to Indian Sign Language, one describes the hand-tracking method used here and one covers interface design principles. Section 6.1 compares them and Section 6.2 gives the insights for this project.

6.1 Comparison of approaches, findings and limitations

Paper	Approach	Key findings	Limitations	Relevance to this project
[1] Bragg et al. (2019), *Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective*, ASSETS '19	Report of a workshop of vision, linguistics, Deaf-culture and HCI experts; reviews the state of the art	Progress is held back by small datasets, little Deaf involvement and little attention to how output reaches users; calls for interdisciplinary, user-centred work	Position paper with no new system or measurements; mainly about ASL	Motivates designing confirmation, correction and uncertainty into the interface (gap 1)
[2] Adeyanju, Bello and Adegboye (2021), *Machine learning methods for sign language recognition: A critical review and analysis*, Intelligent Systems with Applications	Bibliometric and technical review of 649 Scopus publications on vision-based recognition	Feature extraction followed by an ML classifier is the dominant approach; reliable recognition is "still an open problem"	Scopus only; compares methods, not user experience	Supports the landmark-features plus classifier design for static signs
[3] Sridhar et al. (2020), *INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition*, ACM Multimedia	4,287 videos of 263 ISL word signs by Deaf students; pose keypoints with deep models	94.5% accuracy on INCLUDE-50 and 85.6% on all 263 words	Word signs need video and motion models; recorded in one school	Shows the next step after static signs and why this project limits itself to fingerspelling (gap 3)
[4] Velmathi G and Goyal (2023), *Indian Sign Language Recognition Using Mediapipe Holistic*, arXiv	MediaPipe Holistic keypoints with CNN and LSTM models	CNN better for static ISL letters; LSTM better for gesture-based phrases	Preprint; no interface or user evaluation	Supports MediaPipe keypoints for static ISL letters

[5] Zhang et al. (2020), *MediaPipe Hands: On-device Real-time Hand Tracking*, arXiv / CVPR 2020 workshop	Palm detector plus hand-landmark model predicting 21 keypoints from one RGB camera	Real-time on mobile devices with high prediction quality	Harder with overlapping hands, blur and poor light	The landmark extractor used in training and in the browser; enables on-device privacy
[6] Shneiderman et al. (2016), *Designing the User Interface*, 6th ed.	Design principles, including the Eight Golden Rules	Consistency, shortcuts, feedback, closure, error prevention, reversal, user control and low memory load improve usability	Expert guidelines; must be checked with users	Used to design and review every screen (Section 12.7)

6.2 Comparison and insights

The recognition papers [2], [3], [4] measure success by accuracy. Word-level ISL recognition [3] needs video models and large recorded datasets. Static letters can be recognised from one frame of hand keypoints [4], [5], which is light enough to run in a browser. Paper [1] points out what the accuracy-focused work leaves out: how users see, trust and fix the output, and the fact that datasets come from few signers, so real-world accuracy is lower than reported. This project therefore pairs a lightweight landmark classifier with an interface built around confidence, uncertainty and correction, designed with the rules in [6].

7. Research Gaps

Each gap below is supported by the literature in Section 6 and by the user evidence in Section 4.4.

No.	Research gap	Literature evidence	User evidence	How this project responds
1	Recognition systems report accuracy, but not how results are shown to users or how wrong results are handled	[2], [3], [4] judge success by accuracy; [1] calls for user-centred design	F3, F5: correction is the top feature; nobody wants a silent guess	Large result with reference photo, confidence and top-3; uncertain state below 60%; Confirm / Incorrect with correction; Undo
2	Recognition usually needs servers or heavy models, which raises privacy and connectivity barriers	[5] shows on-device hand tracking is possible; [3] uses heavy video models	F4: privacy and camera control, real-time results	Everything runs in the browser (WebAssembly + 0.46 MB MLP); no frames are uploaded or stored
3	Word-level ISL needs motion models and large video datasets, which are not yet practical in a browser	[3] word signs need video; [4] static signs work from single frames	F1, F6: frequent contact, little prior tool use; learners need practice	Scope limited to static fingerspelling A–Z, 1–9, which covers names, numbers and spellings; word signs are future work
4	Reported accuracy is optimistic because datasets come from few signers in controlled settings	[1] dataset limitations; frame leakage in random splits (Section 10)	F7, F8: usefulness only 3.0/5; the fluent signer wants accuracy	Blocked train/test split; leaky random-split figure reported separately; confidence threshold and correction flow in the UI

8. Requirements

The requirements below follow from the findings in Section 4.4. The last column names the finding each requirement responds to.

8.1 Functional requirements

No.	Functional requirement	Source
1	Start and stop the camera only when the user asks; handle denied, missing and busy cameras with a clear message.	F4
2	Detect up to two hands and recognise the 35 static ISL signs A–Z and 1–9 live from the camera.	F1, F6
3	Show the recognised sign in large text with a reference photo, the confidence and the next most likely signs.	F2
4	When confidence is below 60%, show an uncertain state with the closest guesses instead of a single answer.	F5
5	Let the user confirm a result, mark it incorrect with the intended sign, retry, or hear it spoken.	F2, F3
6	Save confirmed and corrected results to a searchable, filterable history; allow deleting one or all entries with Undo.	F3
7	Show every supported sign with a reference photo and let the user practise a chosen sign.	F6
8	When unsure, let the user type the intended message and have it spoken.	F5
9	Provide a demo mode that works without a camera.	F6

8.2 Non-functional requirements

No.	Area	Non-functional requirement	Source
1	Privacy	No camera frame leaves the device; only label, confidence, time and feedback are stored locally.	F4
2	Performance	Detection about 12 times per second; a held sign locks in about one second.	F4
3	Offline	All models and WebAssembly files are served with the app; no external service is called.	F4
4	Accessibility	Keyboard operable, ARIA labels and live regions, large-text and reduced-motion settings, dark mode, colour never the only cue.	F2, F4
5	Efficiency	Single-key shortcuts for every frequent action.	F4
6	Reproducibility	The model can be retrained from public data with two scripts; the browser maths is checked against scikit-learn.	Project team

9. System Architecture

SignBridge is a single-page web application with no server-side code. A static web server delivers the app, the hand-landmark model (7.8 MB), the WebAssembly runtime and the classifier weights (0.46 MB). After that, everything runs on the user's device.

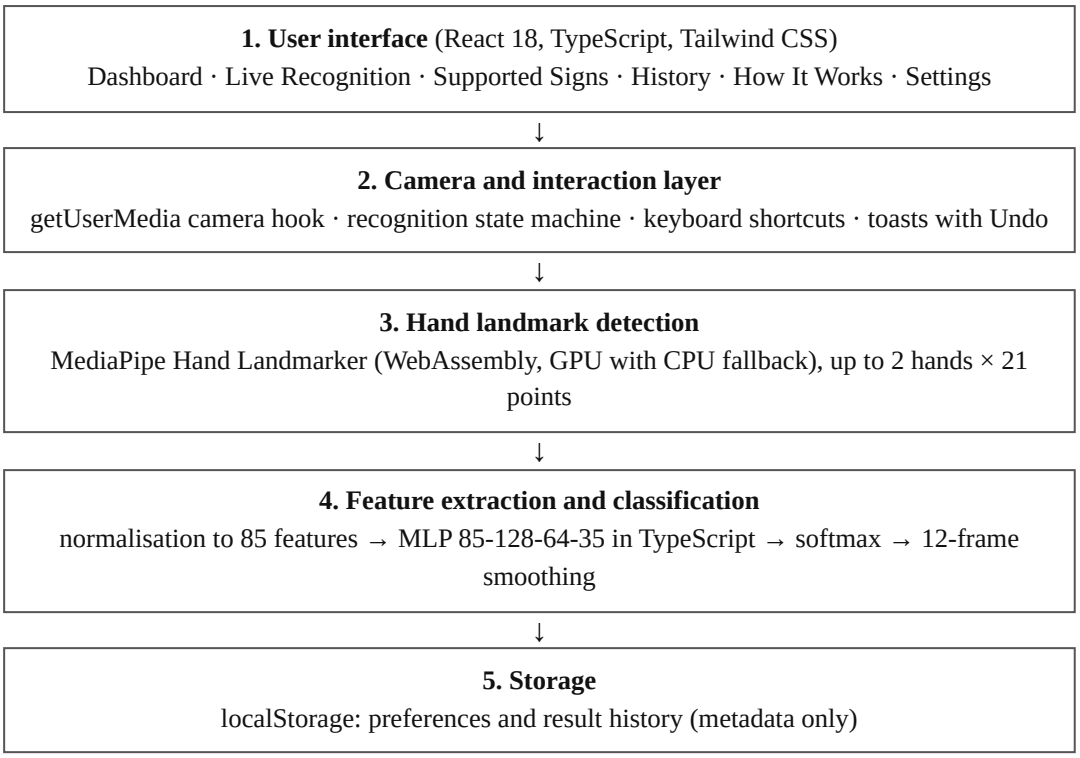


Figure 9.1: layers of the system.

9.1 Recognition flow

- The user starts the camera and presses Start Recognition (or Space).
- About 12 times per second, the Hand Landmarker returns the 21 landmarks of each visible hand, which are drawn over the mirrored video.
- Hands are ordered left to right, centred on the mean of all points and scaled by the largest distance from that centre, giving 84 coordinates plus a two-hand flag.
- The MLP returns a probability for each of the 35 signs. The last 12 predictions are kept. When 9 of them agree and the averaged confidence is at least 60%, the sign is locked and shown as a result.
- Each pose is also compared with 560 training prototypes; poses far from all of them are marked unknown and cannot lock (Section 10.6).
- The user can press Capture now (Enter), and after 3.5 s without a lock the app stops by itself. If confidence is below 60% or the pose is unknown, the uncertain state is shown with the closest guesses.

9.2 Main source files

File	Role
`src/ml/model.ts`	Loads MediaPipe and the weights; detects hands in a video frame

`src/ml/classifier.ts`	Feature normalisation and MLP forward pass (identical to `ml/train.py`)
`src/hooks/useSignRecognizer.ts`	Detection loop, smoothing, locking, landmark overlay
`src/pages/LiveRecognition.tsx`	Main screen: modes, live read-out, result actions, shortcuts
`src/hooks/useShortcuts.ts`, `src/components/Toast.tsx`	Keyboard shortcuts; notifications with Undo
`ml/extract_landmarks.py`, `ml/train.py`	Landmark extraction and model training/evaluation/export
`scripts/parity-check.mjs`	Checks browser predictions equal scikit-learn on 20 test rows

10. Dataset and Model Training

10.1 Dataset

Item	Detail
Source	Indian Sign Language image dataset, Hugging Face `Hemg/Indian_sign_language_dataset` [7]
Images	42,745 RGB images, 128 × 128 pixels, dark background, real hands
Classes	35: digits 1–9 and letters A–Z (about 1,200 images each)
Cleaning	358 files named "*n* copy.jpg" (re-saved duplicates) removed → 42,387 images
Hands detected	98.79% of images (lowest: S 82.8%, Q 84.5%, C 93.9%); images without hands were excluded
One vs. two hands	Digits and C, I, L, O, V are one-handed; A, D, E, F, K, M, T, W, X, Z are two-handed in over 98% of images

10.2 Features

Each image was upscaled to 256 × 256 and passed to MediaPipe Hand Landmarker (up to two hands). Hands are sorted left to right by their mean x coordinate. All landmarks are centred on their joint mean and divided by the largest distance from it, which removes the effect of hand position and distance from the camera. The 2 × 21 × (x, y) coordinates (zeros for a missing second hand) plus a flag for "two hands" give 85 features. To support left-handed signers and mirrored cameras, every training sample was also added in horizontally mirrored form.

10.3 Train/test split

The images are consecutive frames of a few recordings per class, so neighbouring frames are almost identical. A random 80/20 split would put near-copies of test images into training. Instead, for each class the images were ordered by file number and the **last 20% were held out** (blocked split): 33,496 training images (66,992 with mirroring) and 8,377 test images. For comparison, the same MLP on a random split scores 99.95%, which shows how much a leaky split inflates the result.

10.4 Models compared

Model	Configuration	Test accuracy	Macro-F1	Mirrored test acc.	Train time	Inference / sample
kNN	k = 5	97.79%	97.32%	97.79%	< 0.1 s	0.206 ms
Random Forest	300 trees	97.88%	97.46%	97.84%	63.8 s	0.020 ms
SVM	RBF kernel, C = 10	97.88%	97.43%	97.88%	3.0 s	0.290 ms
MLP (chosen)	85-128-64-35, ReLU, Adam, early stopping	**97.86%**	**97.42%**	97.86%	10.7 s	0.001 ms

All four models perform within 0.1 percentage points of each other, which shows that the landmark features carry most of the information. The MLP was chosen because its weights are a small JSON file (0.46 MB) and its forward pass is a few lines of TypeScript, with no ML library needed in the browser. The parity check confirms that the browser gives the same top class as scikit-learn on 20 held-out rows (largest probability difference 4.4×10^{-7}).

10.5 Per-class results and errors

34 of 35 classes have an F1 of 0.99 or higher. The one weakness is **digit 9**: 172 of its 240 test images were predicted as **S** (recall 26.7%), which also lowers the precision of S (53.5%). In the held-out recording the hand shape for 9 looks very much like S in landmark space. More varied training images of 9 and S are the clearest next improvement. In the app, the confidence threshold and the "other candidates" line make this kind of confusion visible to the user.

10.6 Unknown-sign detection

A softmax classifier always assigns one of the 35 classes, often with near-100% probability, even to a hand shape it has never seen. Early live tests showed exactly this: random gestures were labelled as random letters. We therefore added a second check (`ml/ood.py`). For each class, k-means picks 16 prototype poses from the training features (560 in total). A live pose whose distance to the nearest prototype exceeds a threshold is treated as *unknown*: its confidence is capped at 30%, so it can never lock, and the user sees "not a sign shape the model knows" with the closest guesses. The threshold is 1.25 × the 95th-percentile distance of the held-out test set (0.70). It keeps 95.6% of real held-out signs, and rejects 99.3% of synthetic non-signs made by combining hands from two different classes and 99.6% of real signs with randomly bent fingers. In addition, if no sign has locked after 3.5 seconds, the app shows the uncertain state with the closest guesses instead of waiting.

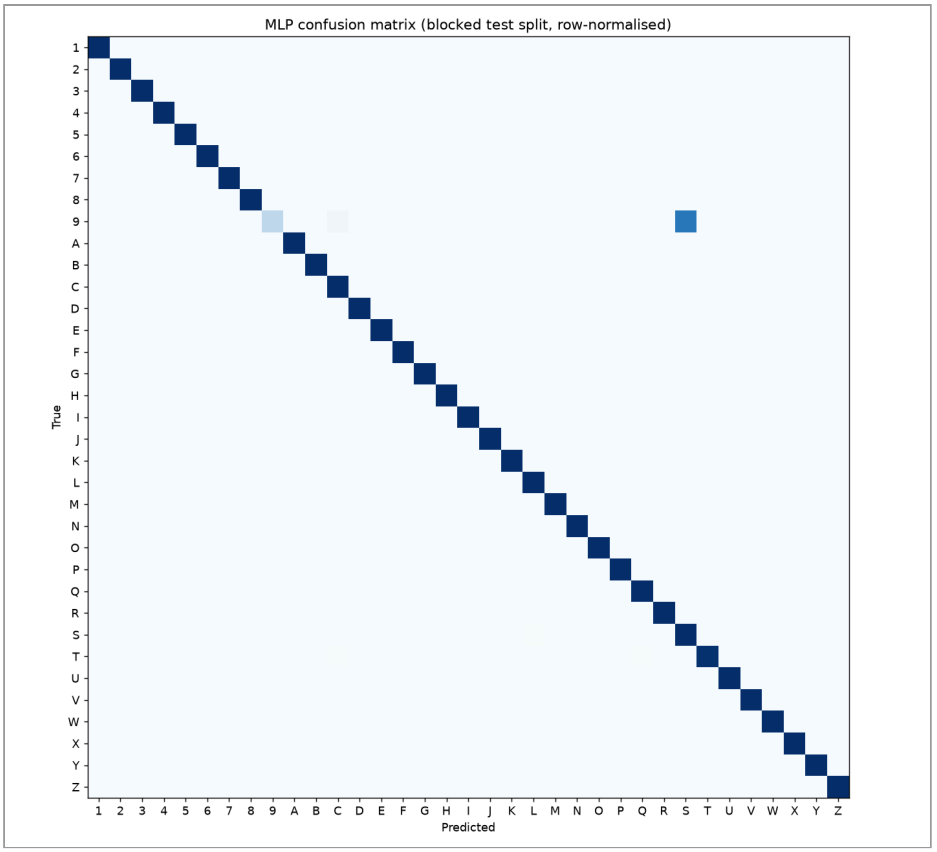


Figure 10.1: MLP confusion matrix on the blocked test split (row-normalised). The off-diagonal cell is 9 → S.

11. Use Cases

11.1 Use-case summary

No.	Use case	Summary	Task	Main users
1	Recognise a sign	Start the camera, hold a sign, read the result in text and speech	T1	ISL users, hearing partners
2	Correct or undo a result	Mark a result incorrect, choose the intended sign, or undo a save	T2	All
3	Practise a sign	Pick a sign in Supported Signs, practise it on camera and see if it matches	T3	Learners
4	Review history	Search, filter and delete past results	T4	All
5	Recover from problems	Camera denied or missing, model not loaded, low confidence	T5	All

11.2 Use case 1: Recognise a sign

Step	Actor / system action
1	The user opens Live Recognition. The model loads in the background and the badge shows "Model ready".
2	The user presses Start Recognition (Space). If the camera is off, the browser asks for permission and the camera starts.

3	The system draws the hand skeleton, shows how many hands are detected and the current guess with a confidence bar.
4	The user holds the sign still. When 9 of the last 12 frames agree at 60% or more, the system locks the result.
5	The system shows the sign in large text, a reference photo, the confidence and other candidates.
6	The user confirms (C), marks it incorrect (X), hears it (S) or tries again (R). A notice confirms the save, with Undo (U).

Precondition: a camera is available and permission is granted. **Alternate flows:** camera denied or missing → message with Retry and Demo mode; confidence below 60% on Capture → uncertain state with closest guesses. **Postcondition:** the result is saved with its feedback, or nothing is saved.

12. Wireframes and Interaction Design

12.1 Purpose and approach

Low-fidelity wireframes were drawn for the five key screens before they were built, so that layout, order of information and interaction could be checked against the user needs from Sections 4 and 5. They are drawn in grey only, so that attention stays on structure rather than colour. Numbered markers link each element to the need it serves, and each wireframe is shown with the matching screen of the working prototype.

12.2 Key user tasks

Task	User task	Findings	Screens
T1	Get one fingerspelled sign recognised and shown as text and speech	F1, F2	W1, W2, W3
T2	Notice when the result is uncertain or wrong and fix or undo it	F5, F3	W3, W5
T3	Practise a chosen sign against a reference	F6	W4, W2
T4	Find, review and delete earlier results	F3	W5
T5	Recover from camera or model problems	F4	W2

12.3 Interaction flow

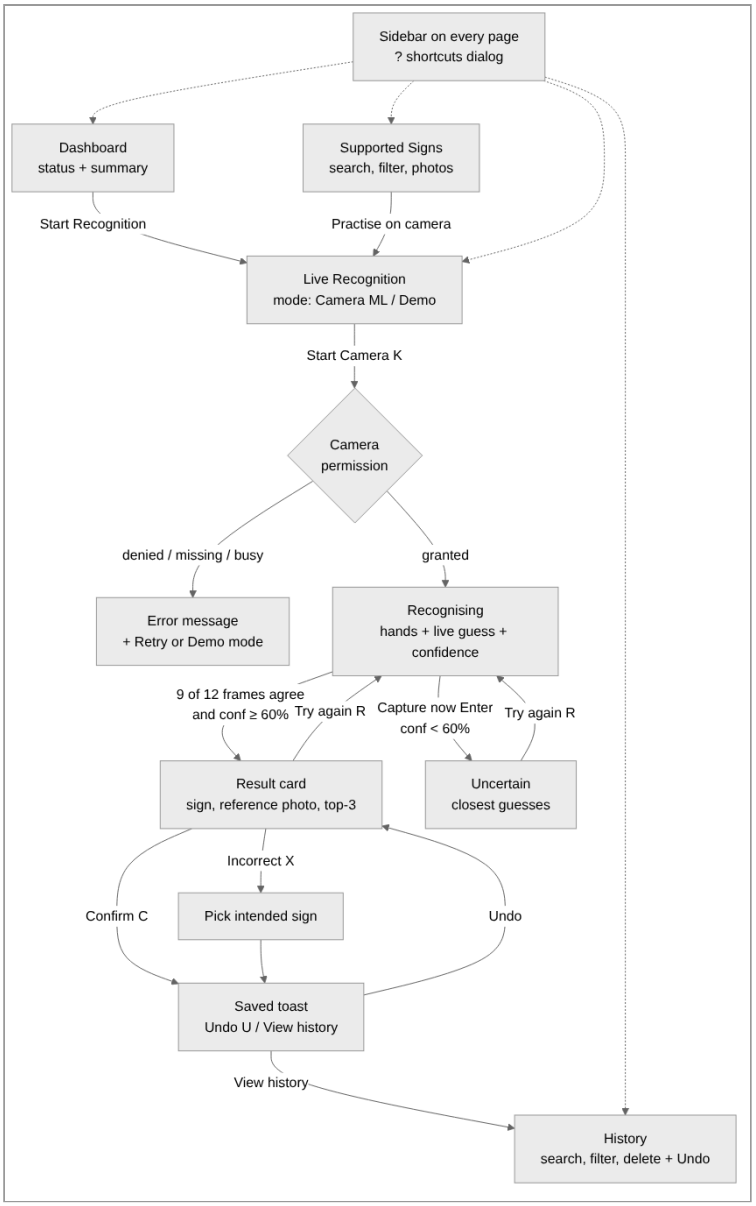
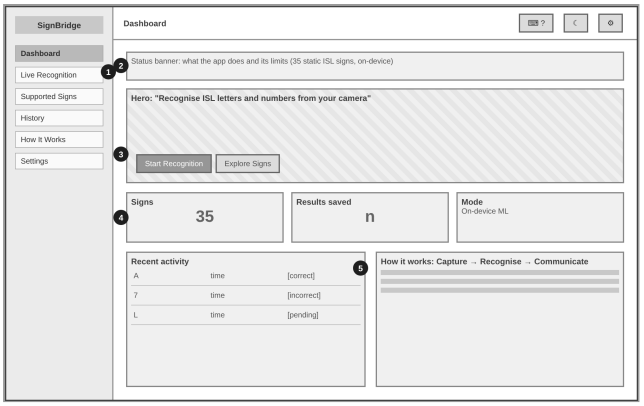


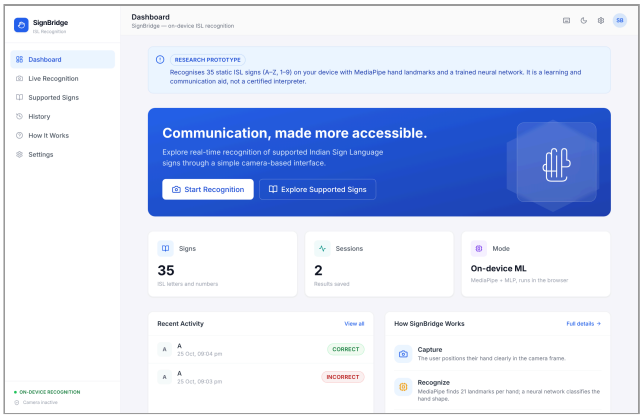
Figure 12.1: interaction flow between the screens of the prototype.

12.4 Wireframes and matching prototype screens

W1 Dashboard



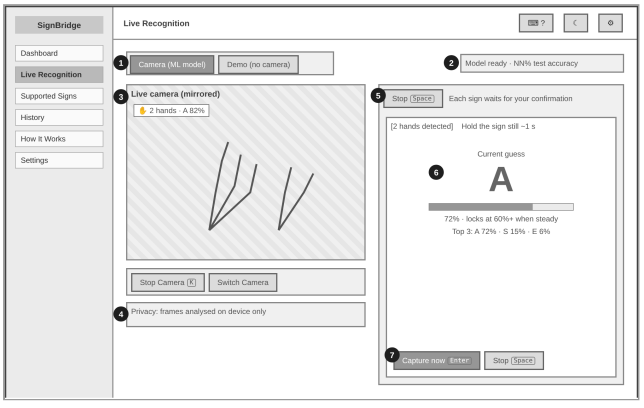
Wireframe W1



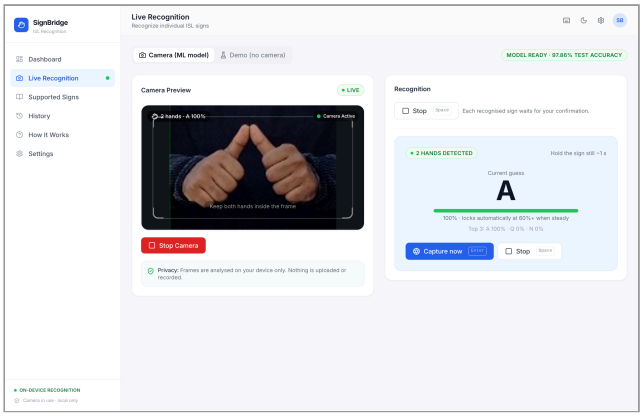
Prototype

No.	Element	Why it is there (user need)	Need
1	Sidebar navigation	The same six links on every page, so users always know where they are and can go anywhere in one click	F4
2	Status banner	Says in plain words what the app can and cannot do (35 static signs, not an interpreter)	F7
3	Hero with Start Recognition	One clear entry to the main task T1	F4
4	Summary cards and recent activity	Shows the vocabulary size and recent results without opening History	F3
5	How it works	Three-step explanation for first-time users	F6

W2 Live Recognition (recognising)



Wireframe W2

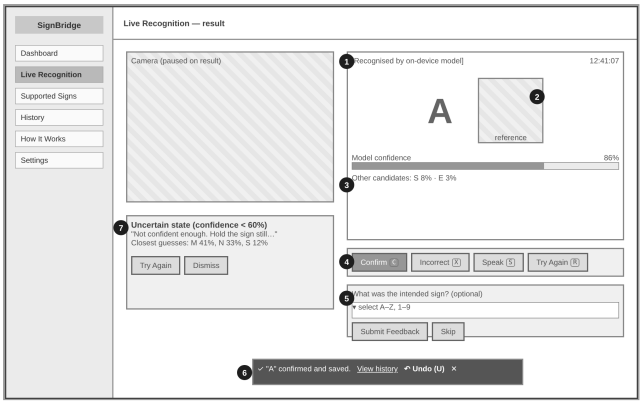


Prototype

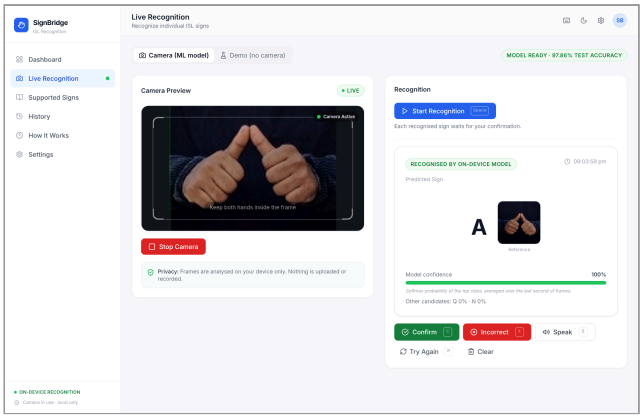
No.	Element	Why it is there (user need)	Need
1	Mode switch: Camera (ML) / Demo	Works with or without a camera	F6
2	Model status badge	Shows whether the model is ready and its tested accuracy	F7
3	Mirrored camera with hand skeleton and chip	Users see that their hands are found, like a mirror	F2
4	Privacy note	States that frames stay on the device	F4

5	Start/Stop with key hint	One action that starts and stops recognition; Space shortcut	F4
6	Live guess, confidence bar, top 3	Continuous feedback while the sign is held	F2, F5
7	Capture now / Stop	The user decides when to take the result	F3

W3 Result, correction and undo



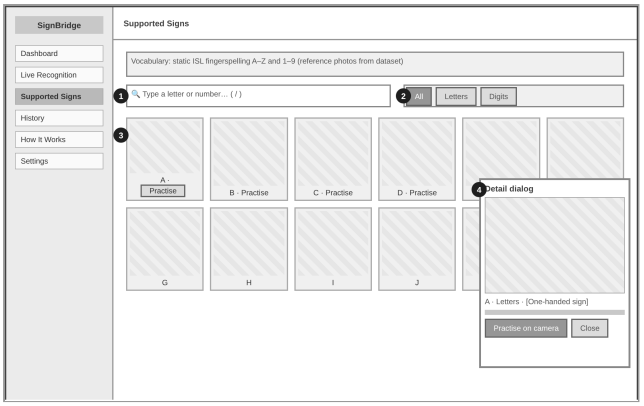
Wireframe W3



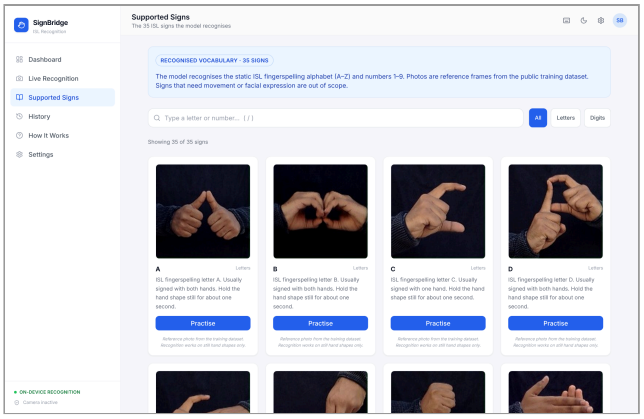
Prototype

No.	Element	Why it is there (user need)	Need
1	Mode badge and time	Says the result came from the model and when	F7
2	Large label and reference photo	Readable from a distance; the photo lets both people check the hand shape	F2
3	Confidence and other candidates	Shows how sure the model is and what else it might be	F5
4	Confirm / Incorrect / Speak / Try again with key hints	One click or key for each next step; speech for hearing partners	F2, F3
5	Correction form	Records the intended sign when the model was wrong	F3
6	Saved notice with Undo and View history	Closure plus a way back	F3
7	Uncertain state	Below 60% the app does not guess; it shows the closest options	F5

W4 Supported Signs



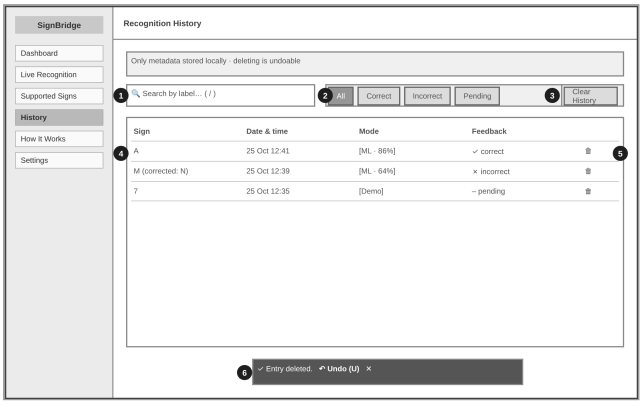
Wireframe W4



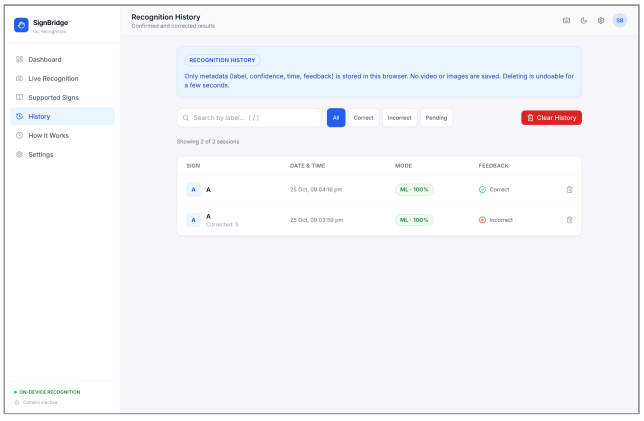
Prototype

No.	Element	Why it is there (user need)	Need
1	Search by letter or number (/)	Find a sign quickly	F6
2	Category filter	Letters or digits only	F6
3	Cards with reference photos	Recognition rather than recall of hand shapes	F6, F2
4	Detail dialog with Practise on camera	One-handed / two-handed note and a direct path to practise (task T3)	F6

W5 History



Wireframe W5



Prototype

No.	Element	Why it is there (user need)	Need
1	Search (/)	Find a past result	F3
2	Feedback filter	Show only correct, incorrect or pending results	F3
3	Clear History (with confirm and Undo)	Remove everything safely	F3, F4
4	Rows with mode, confidence, correction	Review what the model said and what the user corrected	F5
5	Per-entry delete	Remove one mistaken entry	F3
6	Undo notice	Every delete can be reversed	F3

12.5 Design rationale: usability heuristics

Heuristic	How it is applied in the design	Where
Visibility of system status	Model-ready badge, hands-detected badge, skeleton overlay, live confidence, Live camera badge, sidebar camera status	W2, W3
Match with the real world	Mirrored camera; reference photos; plain words ("Hold the sign still ~1 s")	W2, W3, W4
User control and freedom	Stop, Reset, Try again, Undo, Demo mode, auto-save switch	W2, W3, W5
Consistency and standards	Same layout, buttons and colours on every page; one result card for both modes	W1–W5
Error prevention	Confidence threshold, 9-of-12 agreement, confirmation before Clear History	W2, W3, W5
Recognition rather than recall	Reference photos, key hints on buttons, shortcut list (?), correction from a list	W3, W4
Flexibility and efficiency	Single-key shortcuts; Capture now; auto-save mode for continuous spelling	W2, W3
Aesthetic and minimalist design	One main task per screen; secondary actions as ghost buttons	W1–W5
Help users recover from errors	Specific camera error messages with Retry; model error offers Demo mode; uncertain state with tips	W2, W3
Help and documentation	How It Works page, shortcut dialog, inline tips	All

12.6 Design rationale: Shneiderman's Eight Golden Rules [6]

No.	Golden rule	How it is applied in the design	Where	Need
1	Strive for consistency	Same sidebar, header, card layout and button styles on every page. Green always means confident or confirmed, amber uncertain, red error. Demo and camera modes share one result card. Shortcut letters match the action names (C confirm, S speak, R retry).	W1–W5	F4
2	Enable frequent users to use shortcuts	Single-key shortcuts: Space start/stop, Enter capture, C confirm, X incorrect, S speak, R retry, K camera, U undo, / search, ? list. Key hints are printed on the buttons, shortcuts can be turned off in Settings, and they never fire while typing. "Practise" jumps from a sign straight to the camera.	W2, W3, W4, W5	F4
3	Offer informative feedback	Model-ready badge with accuracy; hand skeleton drawn on the video; "2 hands detected"; live guess with confidence bar and top 3; notice after every save.	W2, W3	F2, F5
4	Design dialogs to yield closure	Recognition ends in a locked result card. Confirm or Incorrect ends the task with "'A' confirmed and saved" and a View history link.	W3, W5	F3
5	Offer error prevention and simple error handling	A sign locks only after 9 of 12 frames agree. Below 60% confidence the app shows an uncertain state with the closest guesses instead of a wrong answer. Clear camera messages for denied, missing and busy cameras. If the model fails to load, Demo mode is offered. Clear History asks first.	W2, W3, W5	F5

6	Permit easy reversal of actions	Undo (button or U) after confirming, marking incorrect, deleting one entry and clearing all history. Undo after a save also puts the result back on screen. Stop, Reset and Try again are always visible.	W3, W5	F3
7	Support internal locus of control	The camera starts only on request and recognition only on command. Capture now lets the user decide when to lock. The user chooses manual confirm or auto-save in Settings. Nothing leaves the device.	W2, W3, Settings	F3, F4
8	Reduce short-term memory load	Reference photo next to the result, so the user does not have to remember the hand shape; Supported Signs with photos; ? lists every shortcut; key hints on buttons; recent results on the Dashboard.	W1, W3, W4	F6, F2

13. Working Prototype

The prototype is a working web application, not a mock-up: it uses the real camera and the trained model. To run it, install Node.js 18 or newer, run `npm install` and `npm run dev` in the project folder, and open http://localhost:5173 in Chrome, Edge or Safari. A production build (`npm run build`) can be served from any static web host.

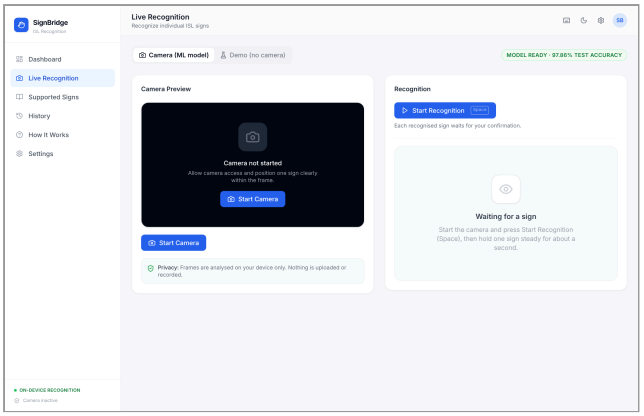


Figure 13.1: Live Recognition, camera on, model ready.

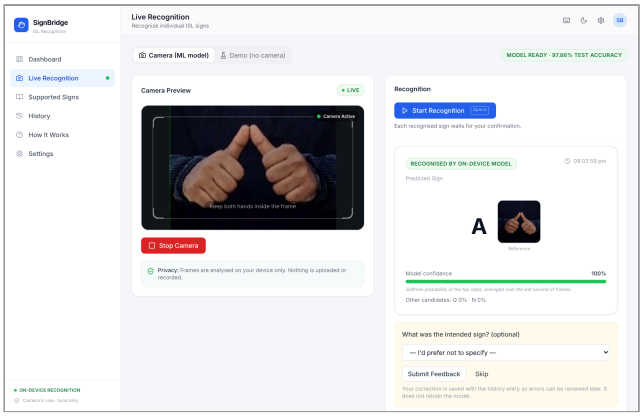


Figure 13.2: marking a result incorrect and choosing the intended sign.

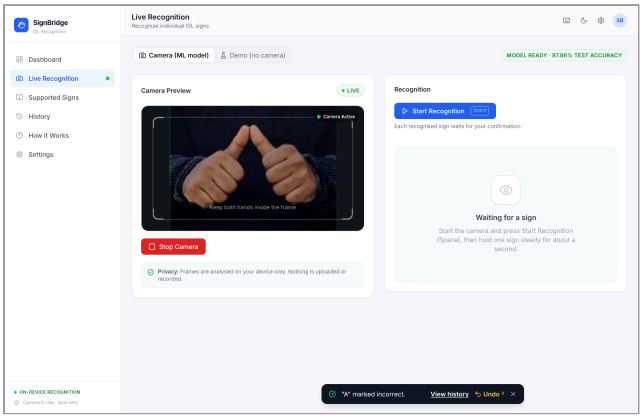


Figure 13.3: saved notice with View history and Undo.

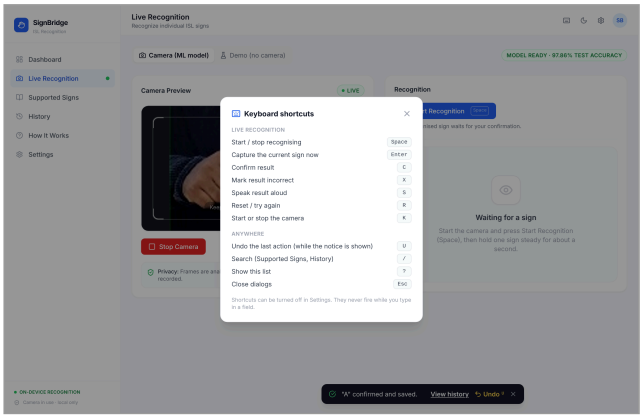


Figure 13.4: keyboard shortcuts dialog (?).

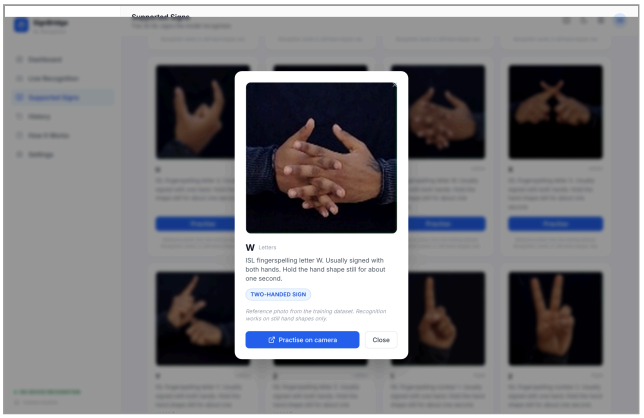


Figure 13.5: sign detail with Practise on camera.

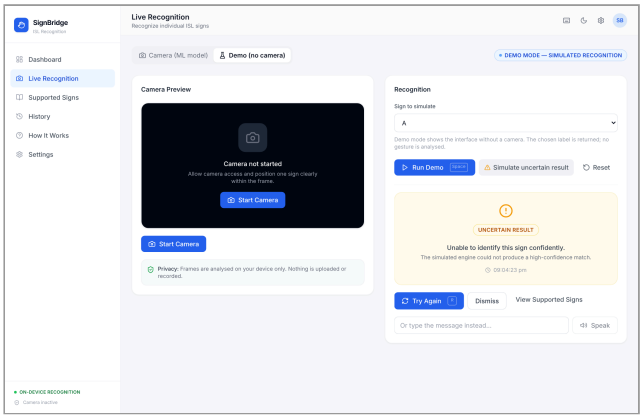


Figure 13.6: uncertain-result state (Demo mode).

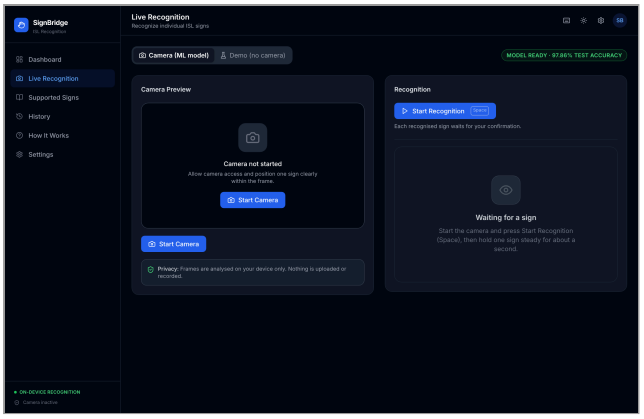


Figure 13.7: dark mode.

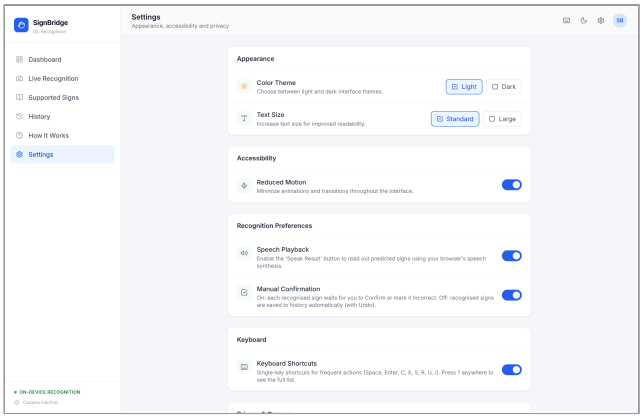


Figure 13.8: Settings (theme, text size, motion, speech, manual confirm, shortcuts, privacy).

13.1 Implementation components

Component	Role
React 18, TypeScript, Vite	Single-page application and build
Tailwind CSS, lucide-react	Custom styling and icons (no prebuilt UI kit)
MediaPipe Tasks Vision (WebAssembly)	Hand landmark detection in the browser
scikit-learn, NumPy, MediaPipe (Python)	Landmark extraction, training, evaluation and export
Web Speech API	Speaking the result aloud
localStorage	Preferences and history (metadata only)
Playwright	Automated end-to-end test and screenshots with a fake camera

14. Research Poster

The research poster (A1 portrait) summarises the problem, user groups, method, personas, findings, literature insights, research gaps, system, ML results, prototype screens and the Eight Golden Rules. The full-resolution file is `docs/poster/SignBridge_A1_Poster.pdf`.

IIIT PUNE / HUMAN–COMPUTER INTERACTION + MACHINE LEARNING

SignBridge

Real-Time Indian Sign Language Recognition in the Browser

A user-centred HCI study and on-device ML prototype

Bittu Mandal (112415048) · Chinmay Ghule (112415054) · Dev Krishan (112415057) · Himanshu Raj (112415082) — IIIT Pune

01 PROBLEM

- Few hearing people know ISL. Names, numbers and spellings are hard to share without an interpreter.
- Accuracy alone misses how results are shown and how users recover when the system is wrong.

02 USER GROUPS

PRIMARY

ISL users · Hearing communication partners: family, staff and classmates

SECONDARY ISL learners · Technical evaluators

03 RESEARCH METHOD

10-question survey · 10 respondents · 6 literature sources

User survey → Findings → 3 personas → Literature review → Research gaps → Wire-frames → Working prototype → Testing

Survey → design requirements → tested prototype

04 PERSONAS

Survey-derived composites; names and ages are illustrative.

Meera, 24
Deaf ISL signer
Names and numbers without an interpreter; apps need reliability.

Rohit, 21
Hearing classmate; no ISL
Real-time text + speech, with a way to fix wrong results.

Sana, 20
ISL learner
Speech, privacy, and typing when a sign is not recognised.

05 KEY FINDINGS

SURVEY · n = 10

- **8/10** rate communication very or extremely difficult.
- **7/10 want speech; 7/10 want text** (5 want both).
- **Correction** is the most-wanted feature (3/10).
- When unsure: **4 ask to repeat; 3 want to type.** Show uncertainty and offer a recovery path.
- Mean usefulness: **3.0/5.** Reliability must be earned.

06 LITERATURE INSIGHTS

- **Design beyond accuracy.** Involve Deaf users. Bragg et al., ASSETS 2019
- **Word signs need temporal modelling.** INCLUDE uses video. Sridhar et al., ACM MM 2020
- **Static letters suit hand keypoints.** On-device tracking enables a lightweight pipeline. Zhang et al. 2020; Velmathi & Goyal 2023

07 RESEARCH GAPS

- Results need confidence and a clear correction path.
- Server recognition adds privacy and internet barriers.
- Word-level video models exceed this prototype's scope.
- Controlled datasets and frame leakage inflate accuracy.

08 SYSTEM

Camera (User-controlled) → MediaPipe (2 hands × 21 points) → Features (85 normalised) → MLP (85-128-64-35) → Smoothing (12 frames; ≥60%) → Output (Result + speech)

Locks when 9 of 12 frames agree and confidence ≥60%.

Runs 100% in the browser — no video leaves the device

09 ML RESULTS

97.86%
MLP · BLOCKED TEST ACCURACY

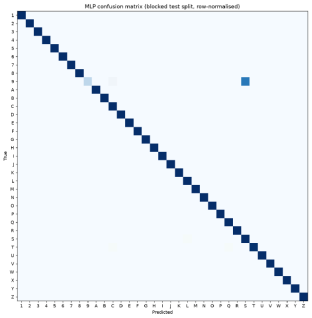
35 classes
STATIC FINGERSPELLING · A–Z, 1–9

42,745 source images · Blocked test: 8,377 images

Model	Accuracy
kNN	97.79%
Random Forest	97.88%
SVM	97.88%
MLP (chosen)	**97.86%**

Chosen for deployment
0.46 MB model; inference in plain TypeScript.

Split matters
Random split: **99.95%**. Near-duplicate frames inflate the result.

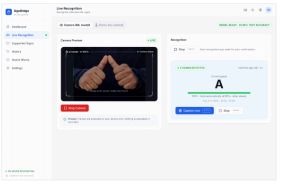


Main error: 9 → S
172 held-out frames confused.

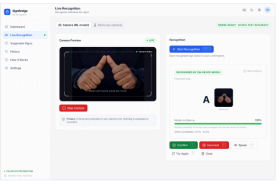
Last 20% of frames per class held out. Generalisation to new signers and backgrounds still needs live evaluation.

10 PROTOTYPE

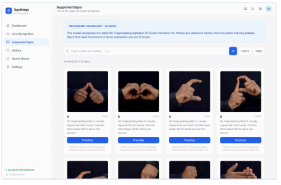
Original interface captures and low-fidelity design



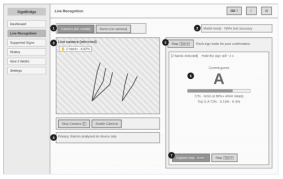
Live recognition with hand skeleton and live confidence



Result with reference photo, confirm / incorrect / undo



Supported signs with reference photos



Low-fidelity wireframe (W2)

UI demonstrations use a recorded test feed. They are not a live-user evaluation.

11 SHNEIDERMAN'S 8 GOLDEN RULES

- **Consistency** · Same layout and colours
- **Shortcuts** · Space, C, X, S, R, U, ?
- **Informative feedback** · Hand skeleton + live confidence
- **Closure** · "Saved" notice + View history
- **Error prevention** · 60% threshold; uncertain state
- **Easy reversal** · Undo everywhere
- **User control** · Camera + recognition on command; auto-save toggle
- **Low memory load** · Reference photo beside every result

12 CONCLUSION & FUTURE WORK

Real-time, private ISL fingerspelling recognition, with an interface built around uncertainty and correction.

NEXT

Data · More signers and backgrounds

Models · Word-level signs (sequence models)

Evaluation · Study with Deaf ISL users



Interaction flow

REFERENCES Bragg et al. 2019; Adeyanju et al. 2021; Sridhar et al. 2020; Velmathi & Goyal 2023; Zhang et al. 2020; Shneiderman et al. 2016.

Figure 14.1: research poster.

15. Testing and Results

15.1 Model evaluation

On the blocked held-out split (8,377 images, 35 classes) the MLP reached 97.86% accuracy and 97.42% macro-F1, and the same score on mirrored test images (Section 10). The main error is 9 predicted as S.

15.2 Functional and end-to-end tests

No.	Test	Expected result	Result
1	TypeScript type check and production build	No errors	Pass
2	Parity check: browser MLP vs. scikit-learn on 20 held-out rows	Same top class	Pass (20/20, max difference 4.4×10^{-7})
3	Automated browser test with a fake camera showing a held-out "L" image: Space → result → C → U	"L" recognised, saved, then removed by Undo	Pass
4	Two-hand sign "A" from a held-out image	Two hands detected, skeleton drawn, "A" locked	Pass
4b	Non-sign pose (held-out "L" rotated 90°)	Marked unknown, uncertain state with closest guesses, no wrong label	Pass
5	Camera permission denied / no camera	Clear message with Retry; Demo mode still usable	Pass
6	Shortcuts while typing in the search field	No shortcut fires	Pass
7	Delete one history entry, then Undo	Entry restored	Pass
8	Responsive layout at 390 px width; dark mode	Usable layout, readable contrast	Pass

16. Conclusion and Future Scope

SignBridge shows that static ISL fingerspelling can be recognised in real time entirely inside a web browser. It combines MediaPipe hand landmarks with a small neural network that reaches 97.86% accuracy on held-out data. The HCI work produced an interface that treats the model as fallible: it shows confidence and alternatives, refuses to guess below a threshold, and makes every action reversible. All eight of Shneiderman's Golden Rules are implemented and mapped to specific screens.

Limitations. The user survey is small (10 respondents, a convenience sample, mostly hearing people) and some self-reported group labels are inconsistent with the stated ISL familiarity, so the findings guide the design but are not statistically representative. The dataset contains few signers against one dark background, so accuracy on new people, skin tones, lighting and backgrounds is expected to be lower. Digit 9 is often confused with S. Only static signs are covered.

Future scope.

- Collect consented data from more signers and backgrounds, and fine-tune with the corrections users already record.

- Add word-level signs with a sequence model (e.g. LSTM or transformer over landmark sequences) trained on INCLUDE [3].
- Join fingerspelled letters into words with a word-completion step.
- Run a usability study with Deaf ISL users and their communication partners.

17. References

[1] D. Bragg, O. Koller, M. Bellard, L. Berke, P. Boudreault, A. Braffort, N. Caselli, M. Huenerfauth, H. Kacorri, T. Verhoef, C. Vogler and M. R. Morris, "Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective," in *Proc. ASSETS '19*, 2019, pp. 16–31. doi:10.1145/3308561.3353774.

[2] I. A. Adeyanju, O. O. Bello and M. A. Adegboye, "Machine learning methods for sign language recognition: A critical review and analysis," *Intelligent Systems with Applications*, vol. 12, 200056, 2021. doi:10.1016/j.iswa.2021.200056.

[3] A. Sridhar, R. G. Ganesan, P. Kumar and M. Khapra, "INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition," in *Proc. 28th ACM Int. Conf. on Multimedia*, 2020. doi:10.1145/3394171.3413528.

[4] G. Velmathi and K. Goyal, "Indian Sign Language Recognition Using Mediapipe Holistic," arXiv:2304.10256, 2023.

[5] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang and M. Grundmann, "MediaPipe Hands: On-device Real-time Hand Tracking," arXiv:2006.10214, 2020.

[6] B. Shneiderman, C. Plaisant, M. Cohen, S. Jacobs, N. Elmqvist and N. Diakopoulos, *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th ed. Pearson, 2016.

[7] Indian_sign_language_dataset (Hemg), Hugging Face, 42,745 images, 35 classes. https://huggingface.co/datasets/Hemg/Indian_sign_language_dataset